

Unit 10, Lesson 2: Outliers



Benchmarks

MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.



Learning Targets

- ☐ I can determine the implications an outlier has in a given data set.
- ☐ I can relate the impact of an outlier to which measures of center and variation best describe a distribution.



10.2.1 Warm-Up: Precision Is Key

Becky Miller is a quality control engineer with GE Aviation. Like many other careers, manufacturing engineering has many moving parts that make up an entire process. Becky says that manufacturing is problem-solving.

1. List at least two steps that you take when faced with a difficult task.

Becky mentions that an error “about the size of a human hair” can make a difference in manufacturing jet engines.

2. What is a time that you have made a small error that caused a big impact in your next steps?

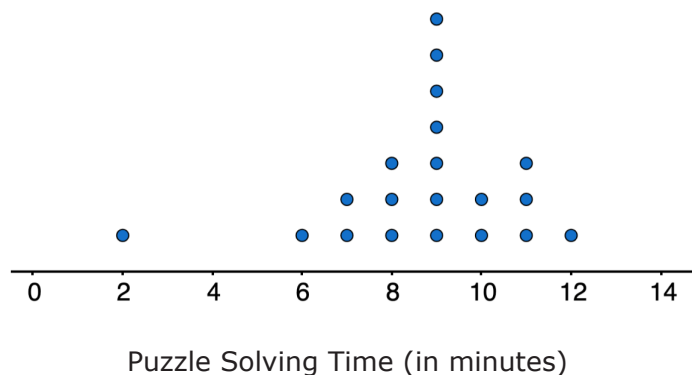


10.2.2 Exploration: Determining, Describing, and Comparing Numerical Data Represented Graphically

Work with your partner to complete the following.

- Twenty students timed how long it took each of them to solve a puzzle. Below is a list of their solution times, in minutes, and a dot plot displaying the data.

2, 6, 7, 7, 8, 8, 8, 9, 9, 9, 9, 9, 9, 10, 10, 11, 11, 11, 12



- Discuss with your partner what you notice about the mean and the median of the data. Summarize your discussion.

- What is the range of the data?

- Find the interquartile range (IQR).

Q1	Q3	IQR

- Describe the distribution of data.
- Determine the mean of the data set. Then, draw an arrow ($\uparrow$) below the number line that points to this value.
- Determine the median of the data set. Then, circle the value on the number line that represents this value.

- Imagine that the fastest time of 2 minutes was removed from the data set. Discuss with your partner what effect, if any, removing this value has on the values of center and variation (mean, median, range, IQR). Summarize your discussion.
- How does the shape of the graph change after 2 is removed?

4. Compare the new values with the original values.

	Original Value	>, <, =	New Value after 2 is Removed
Mean			
Median			
Range			
IQR			

5. Which measure(s) of center will be affected if the outlier is removed?

- ☐ mean
- ☐ median
- ☐ both the mean and median

6. Which measure(s) of variation will be affected if the outlier is removed?

- ☐ range
- ☐ interquartile range
- ☐ both the range and the interquartile range



10.2.3 Bring It Together: Outlier Effects

1. Complete each statement to describe the effect the outlier, 2, has on each measure of center or variation.

- The presence of an outlier does not have a significant effect on the

☐ mean.
☐ median.
- The presence of an outlier will change, often significantly, the value of the

☐ mean.
☐ median.
- The presence of an outlier does not have a significant effect on the

☐ IQR.
☐ range.
- The presence of an outlier will change, often significantly, the value of the

☐ IQR.
☐ range.



When outliers are present or a distribution is skewed, the _____ is the preferred measure of center and the _____ is the preferred measure of variation to describe the distribution.



10.2.4 Guided Practice: Data Outliers

1. Zoe needed exact change to purchase a snack and a drink from the vending machine. She asked her friends how much change, in dollars, they had and listed the results.

0.15, 0.15, 0.20, 0.25, 0.25, 0.30, 0.30, 0.30, 0.35, 0.75, 0.85

- a. What do you notice about the data?
- b. Will the mean be higher, lower, or approximately the same as the median? Explain your reasoning or justify your answer by showing your work.
- c. Describe the impact of removing 0.85 from the data set.

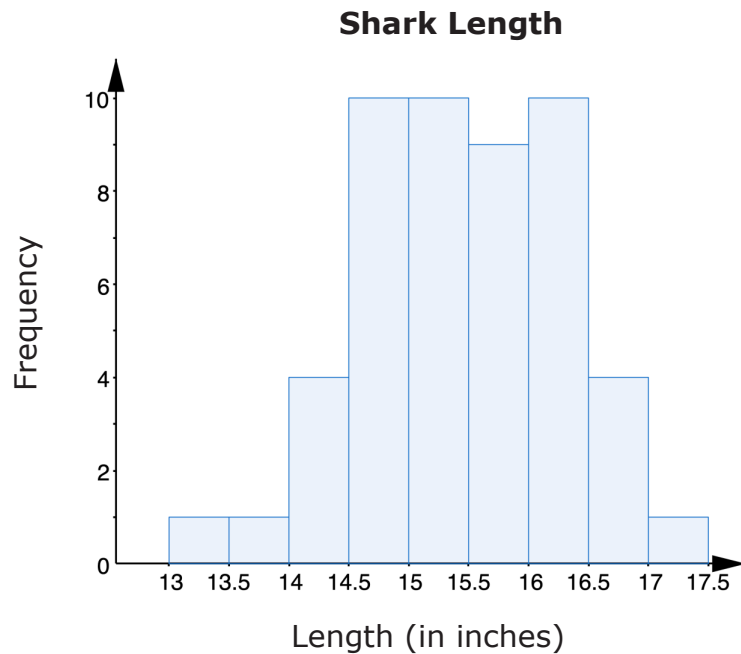
- d. The values 0.85 and 0.75 are outliers in this data set. What happens to the range if 0.85 and 0.75 were both removed from the data set?

- e. The IQR of the full data set is _____.
The IQR with 0.85 and 0.75 removed is _____.
- f. Is there a big difference between the IQR of the original data set and the modified data set?



10.2.5 Exploration: Measures of Center

1. Consider the histogram of the lengths of female sharks.



- Describe the shape of the histogram.
- Circle the value on the horizontal axis of the graph that shows the approximate location of the median of the data.
- Draw an arrow ($\uparrow$) on the graph to indicate the approximate location of the mean of the data.

- d. Compare your answers to B and C with your partner's answers. Discuss your reasoning and, if necessary, make changes.



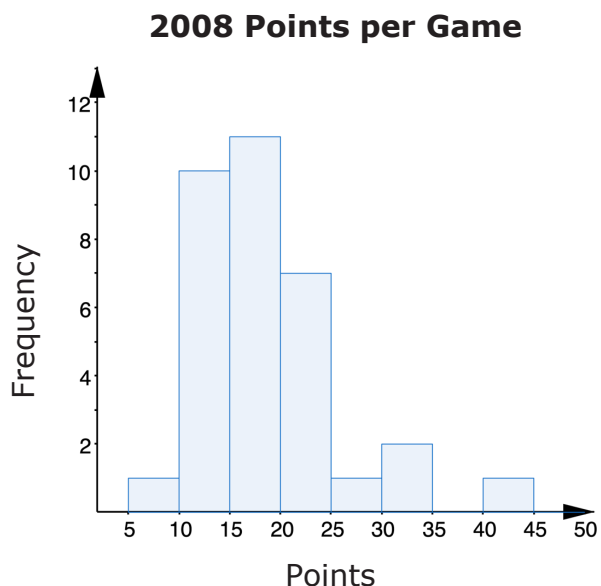
In a symmetrical graph, the mean and the median are located in the middle of the graph, approximately in the same location.

- e. The mean and the median of this histogram are located between 15 – 15.5. Discuss with your partner which measure most accurately describes the center of the histogram. Summarize your discussion.



In a symmetrical graph with no outliers, the _____ is the preferred measure of center and the _____ is the preferred measure of variation to describe the distribution.

2. The histogram displays the number of points per game scored by a college basketball player during the 2008 season.



- a. Describe the shape of the histogram.
- b. Circle the value on the horizontal axis of the graph that shows the approximate location of the median of this data.
- c. Draw an arrow ($\uparrow$) on the graph to indicate the approximate location of the mean of the data.
- d. In a symmetrical graph, the mean is the preferred measure of center to use. The histogram for points scored is skewed. Discuss with your partner why the mean would not be a good representation of the center for points scored. Summarize your discussion.
- e. In this graph, the median is located between 15 – 20 and the mean is in the 20 – 25 interval. Which measure most accurately describes the center of the histogram?
- f. Which measure of variation should be used to describe the distribution?



In a skewed graph or a graph with outliers, the _____ is the preferred measure of center and the _____ is the preferred measure of variation to describe the distribution.



10.2.6 Your Turn!

The following data set shows the number of tickets a movie theater sold in one day for each movie that is playing at the theater.

162	623	76	277	181	64	105
265	82	263	124	167	150	245

1. One of the values in the data set is an outlier. Explain which value you think is the outlier.

2. Complete the statement.

Removing the outlier, 623, causes the mean to

- ☐ increase
- ☐ decrease
- ☐ remain the same

and the range to

- ☐ increase
- ☐ decrease
- ☐ remain the same

because it is much

- ☐ larger
- ☐ smaller

than the other values in the data set.



10.2.7 Wrap-Up: Cool Down

1. The presence of an outlier has a significant effect on which measures of center and spread? Select all that apply.
 - ☐ Mean
 - ☐ Median
 - ☐ Interquartile Range
 - ☐ Range
2. A data set has an outlier that is significantly less than most of the other values.
 - a. Explain how this impacts the measures selected in number 1.
 - b. Which measures of center and variation would you use to best describe the data?